



Siddaganga Institute of Technology, Tumakuru-572 103

(An Autonomous Institution affiliated to VTU, Belagavi, Approved by AICTE, New Delhi)

Seventh Semester B.E. Examinations January - February 2024

Image Processing and Computer Vision

Time: 3 Hours

Max. Marks: 100

Note : 1. Revealing of Identity in any form in the answer book will be treated as malpractice.
2. Answer any five questions choosing one full question from each unit.

Unit - I

- | | M | BL | CO | PO | PSO |
|---|---|----|----|----|-----|
| 01 a) Identify and illustrate the fundamental steps in digital image processing. | 8 | 2 | 1 | 1 | 1 |
| b) Determine the Euclidean, D4 and D8 distances between the diagonally opposite corners of a 256 x 256 image. | 6 | 3 | 1 | 2 | 1 |
| c) Analyze the effect of reduction in spatial resolution in an image. | 6 | 3 | 1 | 2 | 1 |

OR

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|--|---|---|---|---|---|
| 02 a) Identify and explain the various components of an image procession system. | 8 | 2 | 1 | 1 | 1 |
| b) With relevant equations, analyze the illumination reflectance model of an image. | 6 | 3 | 1 | 2 | 1 |
| c) Identify, whether the pixels P and Q in the image given are 4 or 8 or m-connected. If connected, show the paths. Assume $V=\{70,80\}$. | | | | | |

(P)	70	78	80	82	78	77	78	82	78
	83	76	82	83	82	80	82	82	78
	80	82	80	73	80	80	82	82	80
	80	82	83	83	80	82	83	82	82
	87	80	82	82	82	80	82	80	78
	89	82	82	80	82	82	72	73	80(Q)

6	3	1	2	1
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Unit - II

- | | | | | | |
|---|---|---|---|---|---|
| 03 a) Analyze median filter with necessary equations and an example. | 8 | 3 | 2 | 2 | 1 |
| b) Justify the need of image sharpening and mention applications of sharpening. | 6 | 3 | 2 | 2 | 1 |
| c) Analyze how background features are recovered when using Laplacian operator. | 6 | 3 | 2 | 2 | 1 |

OR

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|--|---|---|---|---|---|
| 04 a) Analyze the mean filter with necessary equations and an example. | 8 | 3 | 2 | 2 | 1 |
| b) Apply 3 x 3 median filter to the following image. Comment on the results. | | | | | |

10	11	12	0	14	15
15	255	45	0	23	32
25	23	22	0	22	34
13	12	24	0	12	12

6	3	2	2	1
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|--|---|---|---|---|---|
| c) Discuss briefly the techniques for detecting lines and edges in an image. | 6 | 2 | 2 | 2 | 1 |
|--|---|---|---|---|---|

Unit - III

- | | | | | | |
|---|---|---|---|---|---|
| 05 a) Analyze region feature description with necessary diagrams and equations. | 8 | 3 | 3 | 2 | 1 |
| b) With necessary equations, analyze Fourier descriptors used characterizing co-occurrence matrices of size K x K | 6 | 3 | 3 | 2 | 1 |
| c) Derive expression for Hotelling transform | 6 | 3 | 3 | 2 | 1 |

OR

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|--|---|---|---|---|---|
| 06 a) Analyze how chain codes are used to represent a boundary with an example | 8 | 3 | 3 | 2 | 1 |
| b) With necessary equations, analyze the basic boundary descriptors. | 6 | 3 | 3 | 2 | 1 |

- c) With necessary equations, analyze how SIFT algorithm finds image locations that are invariant to scale change.

6 3 3 2 1

Unit - IV

- 07** a) With necessary diagram, illustrate the working of pinhole camera.
b) Analyze how depth is computed using stereo vision with necessary geometry and equations.
c) Illustrate 3D video system with block diagram.

8 3 4 2 1

6 3 4 2 1

6 2 4 1 1

OR

- 08** a) With geometry diagram, illustrate the concept of stereopsis along with reconstruction.
b) Illustrate the concept of virtual view synthesis.
c) List the properties of Euclidean geometry.

8 3 4 2 1

6 2 4 1 1

6 2 4 1 1

Unit - V

- 09** a) Differentiate between classification and regression and analyze linear regression with an example.
b) Differentiate between supervised and unsupervised learning with example for each.
c) List some of the attractive attributes of the biological neural network that make it superior to the most sophisticated artificial intelligence.

8 3 5 2 1

6 3 5 2 1

6 2 5 1 1

OR

- 10** a) Analyze the clustering algorithm based on average linkage concept and perform hierarchical clustering using average-linkage algorithm for the following features

	x	y
1	4	4
2	8	4
3	15	8
4	24	4
5	24	12

10 3 5 2 1

- b) Analyze the concept of k-means clustering and develop an algorithm. Perform hierarchical clustering using K-means algorithm for the following features.

	x	y
1	4	4
2	8	4
3	15	8
4	24	4
5	24	12

10 3 5 2 1